

National Aeromodelling Competition

Techfest 2025-26

The competition is launched with the vision to provide a unified national platform for students interested in aerospace and related engineering disciplines to demonstrate their aero-modeling expertise.

Problem Statement:

A team must design, fabricate, and demonstrate a fixed-wing aircraft system that can perform the tasks mentioned in the following rounds with given constraints.

Format of the Competition:

A. Qualifier Round

The aim of the qualifier round is to select top 30 teams to the Main Round. Since the **theme** is **payload capability**, the qualifier round's objective is to select those teams who built aircraft with good payload-carrying capability and can fly safely while carrying the payloads.

The aircraft should carry one or more than one payload (golf balls of weight 45 g and diameter 43 mm—will be supplied by the organizers during the competition) as they can and fly for a minimum time of 30 seconds. The aircraft should then land with the entire payload in it in a designated area. Payload should not come out of the aircraft until touchdown.

The payload should be mounted inside the aircraft and should not be exposed externally. Judges will examine the aircraft and will allow take-off only when they are satisfied with the payload mounting. A maximum time of 3 minutes will be given between the take-off and the landing and the aircraft should complete the round within the time limit.

Score: $\text{Weight of Payload Carried} \times 100 / \text{Weight of Aircraft without Payload}$

Aircraft that are unable to fly for at least **30 seconds** with the payload will receive a **score of 0**. Each team will get **two attempts**, and the **best of the two** will be considered as the team's **final score**. Both attempts must be completed **within 3 minutes**.

Top **30 teams**, based on the score, will qualify for the Main Round from the Qualifier Round. Only aircraft that demonstrate safe flying with payload will be allowed to participate in the main round.

B. Main Round

The team should perform one round trip of payload delivery (golf balls) and return to the point of origin.

The following steps must be performed once the timer starts:

1. Load the aircraft at the point of origin:

- Load the golf balls onto the aircraft while the aircraft is on the ground. The aircraft should be on the ground and not be touched by the team during this step. Only one team member can participate in this step, and the member should load without touching the aircraft. All the opening/closing of the doors/compartments to place the payload should be operated from the transmitter and not manually.
- Cost of the step1 = $C1$ = Time in seconds from the timer starting to the payload compartment closing.

2. Take off:

- Once the aircraft payload compartment doors are closed, the aircraft should take off. The aircraft can be hand-launched in this step to avoid issues of a smooth runway, as the competition is conducted in an open ground.

3. Fly for at least 30 seconds and Land at the point of destination:

- The aircraft can land anywhere on the assigned ground (to make it easier for the participants to achieve a safe landing). The landing point will be considered as the destination.

- Earnings for the trip = $E1 = 10 \times \text{number of golf balls carried}$.
4. Unload the aircraft completely at the destination:
 - Only one team member can participate in this step, and the member should unload the aircraft without touching it. All the opening/closing of the doors/compartments to place the payload should be operated from the transmitter and not manually.
 5. Reload the aircraft at the destination:
 - As done in step 1
 - Cost of the steps 4 & 5 = $C2 = \text{Time in seconds from touchdown to the payload compartment closing after reloading}$.
 6. Take off at the destination. The aircraft can be hand-launched in this step to avoid issues of a smooth runway, as the competition is conducted in an open ground.
 7. Land after a minimum flight time of 30 seconds:
 - The aircraft can land anywhere within the ground (to make it easier for the participants to achieve a safe landing)
 - Earnings for the trip = $E2 = 10 \times \text{number of golf balls carried}$.
 8. Unload the aircraft:
 - As done in step 4
 - Cost of the step 8 = $C3 = \text{Time in seconds from touchdown to the payload compartment closing after unloading}$.

Score = $E1 + E2 - C1 - C2 - C3$

Where

$E1$ = $10 \times (\text{No. of golf balls delivered in Trip 1})$

$E2$ = $10 \times (\text{No. of golf balls delivered in Trip 2})$

$C1$ = Time (sec) from mission start to compartment closed after 1st loading

$C2$ = Time (sec) from touchdown at destination to compartment closed after reloading

$C3$ = Time (sec) from final touchdown to compartment closed after final unloading

A maximum time of **3 minutes** will be given for the entire mission.

Judging:

- 10 points will be awarded for the successful delivery of each golf ball on each trip.
- 1 point will be deducted as cost per second on the ground.
- Net points earned by the team is the score of the team.

Scoring:

- An overall score combining the Qualifier and Final round scores will be calculated.
- Teams with the highest overall score will be declared as winners.
- Overall Score = Normalized Score in Qualifier Round * 0.25 + Normalized Score in Main Round * 0.75.

Design Constraints:

1. The competition requires participants to design and fabricate an RC aircraft. Readymade models, like RTF, ARF, BNF etc., are not allowed.
2. The Thrust-to-Weight ratio must not exceed 1 (without payload). If excess thrust is detected, it will be neutralized either by adding weight below the aircraft at its center of gravity or by adjusting the throttle limits in the transmitter. This adjustment must be completed within 5 minutes.
3. The aircraft should have a maximum weight of 1 Kg (without any payload being added)
4. The propeller diameter should not be greater than 13 inches.
5. The wingspan of the aircraft must not exceed 1.2 meters. No tolerance will be permitted beyond this limit.
6. Only electric motors are allowed. Using IC engines or any other means of providing thrust is prohibited.
7. Use of gyroscopes (gyros) and programming assistance in receivers is prohibited.
8. Programming for any step of the mission is not allowed.
9. Use of FPV or any other support for flying is not allowed.

Team Structure:

1. A minimum of 2 and a maximum of 4 members can be part of a team. An additional mentor can accompany the team but will not be allowed to help them during their participation.
2. A participant can be a member of only one team and is not permitted to operate or fly the aircraft of any other team.
3. The pilot should be registered member of the team or provided by Organizer
4. Team members of a team may be from the same college/school or different ones (school/UG/PG).
5. Any number of teams can participate from one college/school.
6. Only students can participate in this competition.
7. Participants must get a bonafide certificate signed by the respective HoD/Dean/Principal and their school/college ID Card.
8. A safety pilot might be available on the ground (provided by the organizers) and, if needed, have control of your aircraft in case of emergency. The participant can voluntarily take the safety pilot's help or fly independently. However, if the safety pilot feels that the participant's skills are inadequate, the pilot will take over considering the safety aspects. The final decision on this will be that of the safety pilot and judges, and the participant must adhere to the same.

Abstract Submission:

1. All participants must submit an abstract on their aircraft, which should be less than 15 pages (A4 size, 1.5 line spacing) with standard formatting. The Abstract must document the basic design of the aircraft (dimensions, wing areas, velocity, etc.) and explain how the design suits the given problem.
2. Along with the abstract, participants must also send a zip file containing at least 5 and no more than 10 photographs of the aircraft while it is being built.
3. The Abstract must be submitted as per the standard format
4. The shortlisted teams based on the abstract to participate in the zonal rounds would be announced at least 15 days before the competition date.

5. Team will have to choose the slot while submitting the for the round 1 and must present in that time slot if team not able to do so then the will get a chance once all the other participants are done and time permits.

Rules:

1. Each team would be given two attempts in the Qualifier round and two attempts in the payload round. The best score from both attempts will be considered as a score for each round.
2. The timer will start after 30 seconds of the previous team completing their attempt. The participants need to be prepared in time and launch without delay after entering the takeoff zone.
3. The teams would lose their turn if they were not ready in time.
4. The same aircraft should be used in both the rounds and it should be the same one in the abstract. In case of damages during the competition, teams can repair the same aircraft but are not allowed to use a replacement. The repaired aircraft should be ready in time for the turn.
5. If there is a tie, the winner will be decided by a separate round framed by the Judges on the spot. Judges' decisions would be considered final in all cases.
6. Two boundaries, an inner and an outer, will be marked on the ground. The aircraft must land within the inner boundary for the landing to be considered valid. Failure to do so will result in a penalty of 20% of the total points. If the aircraft touches or crosses the outer boundary at any point, the attempt will be disqualified. No tolerance will be permitted in this regard.

General Guidelines:

1. The same aircraft should be used in both rounds, and it should be the same one as in the abstract. In case of damages during the competition, teams can repair the same aircraft but are not allowed to use a replacement. The repaired aircraft should be ready in time for the turn.
2. Teams are suggested to carry additional components (motors, batteries, propellers, etc.) to avoid last-minute surprises at the venue. You will lose time/attempt if you are not ready at your turn.

3. Teams are not allowed to make any modifications to the aircraft after the Thrust-to-Weight ratio has been measured. Any team found doing so will be penalized or may face disqualification.
4. The use of 2.4 GHz radio is required for all aircraft competing. If the participants want to use any other frequency, they must inform the organizers in advance.
5. Receivers installed in the aircraft must be in 'receiver mode only'.
6. Metal propellers are not allowed.
7. Organizers will check all the systems (Servos, motors, etc.) for functionality before the competition. If found not working, teams will be dismissed from the competition.
8. Teams must not share any components of their aircraft (such as motors, ESCs, batteries, or other parts) with other teams. Each team is expected to bring and use its own equipment required for participation in the competition.
9. You are not allowed to change any part of the aircraft (including electronics and battery).
10. The model can have a powered take-off with landing gear or be launched manually by a person standing at ground level.
11. A team member cannot be a part of more than one team.
12. The timer will start 30 seconds after the previous team completes their attempt. The participants must be prepared in time and launch without delay after entering the takeoff zone. The teams would lose their turn if they were not ready in time.
13. In view of stringent safety requirements, if a pilot flies out of the designated flying zone, which includes the overhead of the event organizing and control section, as mentioned at the venue, he/she is disqualified. He/she must immediately turn back and land safely.
14. In case of any disputes or discrepancies, the decision of the organizers shall be final and binding. Any safety violations or misconduct towards the organizers or the test pilot will result in immediate penalties and may lead to disqualification.
15. Changes in rules, if any, will be highlighted on the website and notified to the registered teams.

Timeline:

Abstract Submission Deadline	1st December 2025
Abstract Result Announcement	5th December 2025
Qualifier Round	21st - 22nd December 2025
Main Round	23rd December 2025

Certificate Policy:

1. Top Five teams in the grand finale will be awarded Certificate of Excellence.
2. E-Certificate of participation will be given to the teams scoring more than the critical marks, which will be decided later.

Prize Money:

The Prize money will be awarded to the top 3 Winners via NEFT and will be processed within 30 working days after receiving the Prize Money from Sponsors.

The Winners have to mail the following information (immediately after the announcement of the results) to vaibhav@techfest.org . Subject: Competition name, team id- your position (example- Aeromodelling, Aero-2309234- 1st position) Body of mail

1. Account Holder's Name
2. Account Number
3. Bank name and Branch name
4. IFSC Code
5. Photo of Bank Passbook as proof